

Audio file

[6.4 Myeloproliferative Disorders.mp4](#)

Transcript

The last of the leukemic processes that we're going to discuss are the myeloproliferative disorders. In order to understand the myeloproliferative disorders, we've got to go back to normal hematopoiesis. So recall that in normal hematopoiesis, there is a CD34 positive stem cell, and that that stem cell can produce a myeloid stem cell or a lymphoid stem cell. The myeloid stem cell will eventually become the cells of the myeloid lineage, producing red blood cells, granulocytes, monocytes, and megakaryocytes, and the lymphoid stem cell has the ability to eventually produce the B cells and the T cells. If there is an accumulation of precursors of the myeloid lineage, then we call that acute myelogenous leukemia. If there is an accumulation of cells of the immature lymphoid lineage, then we call that ALL. If there is an accumulation of mature lymphoid cells, for example, T-cells or B-cells, then we would call that chronic leukemia, and there are subtypes that we've already discussed in the previous section. Now, if there is a neoplastic accumulation of mature myeloid cells, then we call that a myeloproliferative disorder. And of course, the myeloproliferative disorders will be subcategorized based on which cell is predominantly produced. So, for example, if the red cells are overproduced, we call that polycythemia vera. If the granulocytes are overproduced, we call that CML, or chronic myelogenous leukemia. If the megakaryocytes are overproduced, or the platelets of the megakaryocytes are overproduced, we call that essential thrombocythemia. And so, this is the general paradigm of the myeloproliferative disorders. Now, I'd like to make one thing very clear before we move forward, and that is that in a myeloproliferative disorder, you actually get neoplastic accumulation of everybody. So, everybody's increased. The red blood cells will be increased, the granulocytes will be increased, the platelets will be increased. However, we name it based on the predominant cell. So, for example, if there's an increase in red blood cells predominantly, we call that P. vera. However, in polycythemia vera, there will also be an increase in neutrophils, basophils, and eosinophils, and there could also be an increase in platelets, for example, as well. This then takes us to your text, which basically states many of the things that I've already mentioned. What is a myelodysplastic syndrome? A neoplastic proliferation of mature cells of the myeloid lineage. It's a disease of late adulthood, and it results in a high white blood cell count with a hypercellular marrow. Now, the white blood cell count is going to go up because there's going to be an increased production of granulocytes, and the granulocytes are counted in the white blood

cell count. So I hope that that's clear. Cells of all of the myeloid lineages will be increased, as I've already mentioned, and it's classified based on the dominant myeloid cell produced, as I've already explained. Generally speaking, there are some complications that are common. There's going to be an increased risk for hyperuricemia and gout. The notion here is that when you're developing a ton of these cells, you're also going to be turning over a ton of these cells. For example, let's say that the patient has polycythemia vera. In that particular syndrome, they overproduce red blood cells. Obviously, every time you produce a red blood cell, you're producing a nucleus, and that nucleus has to be removed. Remember that the red blood cells have their nucleus removed. So every time the nucleus is removed and the nucleus is degraded, that's going to result in activation of the purine degradation pathway, and the end result of the purine degradation pathway is uric acid. So these patients can easily develop hyperuricemia and gout. Same thing with CML. There's going to be an overproduction of granulocytes. Those granulocytes will eventually die, and their nuclear material will need to be turned over, and it's the degradation of the purines that's going to result in hyperuricemia and a risk of gout. Another complication is that they can progress to marrow fibrosis. This is called the spent phase or the burnt out phase. Third complication is that many of the myelodysplastic syndromes can transform to acute leukemia. You've got mutations in a stem cell that allows the stem cell to develop and produce mature cells. However, if those mutations were to progress, you could easily arrest the stem cell in the blast phase, resulting in an acute leukemia. The first of these is CML, or chronic myeloid leukemia. This is a neoplastic proliferation of mature myeloid cells, as I've already explained. However, the predominant cells are granulocytes. One characteristically increased granulocyte are the basophils. And this is very important. If you'll recall from the very first lecture that I gave in this chapter, when we see basophilia, it's highly associated with chronic myeloid leukemia. And you're going to see why this is important in a few minutes as well. If we have a slide and we're looking at chronic myeloid leukemia, what we're going to see is an increase in the granulocytes. So all of these are clearly granulocytes. And we might even see some early precursor granulocytes within the blood. And then characteristically, as we can see off here in the corner, there's also going to be an increase in basophils within the blood, or a basophilia. CML is driven by a 9-22 translocation. And this 9-22 translocation results in a fusion between BCR and ABL. And this fusion results in increased tyrosine kinase activity. The increased tyrosine kinase activity then drives the overproduction of the neoplastic cells. Recall that tyrosine kinase is a signal transducer. It is an oncogene that has the ability to transduce the signal of growth to the nucleus, and so overexpression of tyrosine kinase would result in overgrowth. The first line of treatment is imatinib, and imatinib basically blocks the excessive tyrosine kinase activity. It's a designer drug that has been designed to specifically block this increased tyrosine kinase activity. In these patients, splenomegaly is

common. In fact, you can divide CML into three phases. There's the chronic phase, the accelerated phase, and then finally, the transformation phase, where they develop acute leukemia. Now, in the chronic phase, they have an enlarged spleen, and that's common. If there is an enlarging spleen, that is that the spleen is actually becoming bigger, then that is a marker of the accelerated phase, indicating that the patient's disease is getting worse. And it usually heralds the transformation to an acute leukemia, which would be the worst complication that could arise from CML. transformation to an acute leukemia, when it occurs, can result in either AML or ALL. And this is a very important principle in understanding CML. So I'd like to take a moment to highlight something. Remember that we've got a hematopoietic stem cell, And this hematopoietic stem cell can develop into the myeloid lineage, or it can develop into the lymphoid lineage. And one of the cells of the myeloid lineage are the granulocytes, and it's predominantly the granulocytes that are increased in CML. Now, when you think about where the mutation is in this particular disease, the mutation must be in a stem cell, because the stem cell is the one that has the ability to produce more and more granulocytes, so the mutation could either be here, in that this cell is mutated and is overproducing the granulocytes, or the mutation could be up here. In that this cell is mutated, it's overproducing the myeloid blast, and then the myeloid blast is overproducing the granulocytes. And if I asked you where is the mutation, the answer would be that the mutation is up here. And that is why, when patients actually develop a transformation to acute leukemia, they can develop a transformation where they begin to pile up these cells, which would result in an AML, or they could develop a transformation where they begin to pile up these cells, which would result in an ALL. And that's a very important principle to remember, that patients who have CML can transform to either AML or to ALL. Another high yield discussion concerns distinguishing CML from a leucomoid reaction. This is a very interesting discussion. It's a way that examiners like to play the game. It's a way that you might encounter a patient clinically. So let's briefly take a moment to review. When we look at the blood smear of CML, what we're seeing is a bunch of increased granulocytes and some early precursors of those granulocytes. Now if I were to look at that smear, my differential diagnosis would be acute infection or CML. Acute infection because remember that infection is going to result in the overproduction of granulocytes. And the overproduction of granulocytes would then result in a high white blood cell count. So if I see a high white blood cell count and it's predominantly granulocytes, I can ask the question, is this due to infection, i.e., a leucomoid reaction, or is this due to CML, which would be a neoplastic proliferation? And there's three key ways by which I can make that distinction. The first way I can make that distinction is to look at an enzyme called leukocyte alkaline phosphatase. Leukocyte alkaline phosphatase is an enzyme present in the secondary granules of granulocytes. And leukocyte alkaline phosphatase is an enzyme that's useful for the purposes of inflammation. Now, cells that

are arising due to an inflammatory process, a leucomoid reaction, would be leukocyte alkaline phosphatase positive. And that makes sense because those cells are being generated to fight an infection. However, cells of CML, which are neoplastic, have no intention of fighting infection. They're basically having a party, a neoplastic party, and so they would be LAP negative. So it's very important to know that the CML granulocytes are leukocyte alkaline phosphatase negative, as opposed to those in a reactive condition, like inflammation, in which case the leukocyte alkaline phosphatase would be positive. Another important distinction would be the presence of basophils. If the patient has an infection and gets a high white count with a left shift and has a predominant neutrophilic leukocytosis as a reactive phenomena, the basophils will not go up. The basophils go up in CML, and that's why I've stated a couple times now so far that basophilia is highly associated with CML. The third way that we could make this distinction is to simply look for the translocation. In a reactive leucomoid reaction where we have a high white count with neutrophils due to infection, there would be no 922 translocation. If the 922 translocation is present, it is a defining feature of CMO. The next disease that we want to discuss is polycythemia vera. Now, the idea here is that, again, you have a neoplastic proliferation of mature myeloid cells, especially the red blood cells. However, the granulocytes would be up, raising the white count, and the platelets would also be up. Very important that this particular disease is driven by JAK2 kinase mutation. That's super high yield for examinations, often shows up. JAK2 kinase mutation is what drives polycythemia vera. Clinically, the patients are going to get symptoms of hyperviscosity. Now the idea here is that the blood gets loaded with red blood cells and becomes super thick. And so when it becomes super thick, there's a hyperviscosity to the blood, which then can create certain clinical findings. For example, the patients get blurry vision and headache due to the high hyperviscosity of the blood. There's also an increased risk of venous thrombosis. And can you remind me from chapter one of our discussion, what's a classic venous thrombosis that could arise in these very high yield for examinations, Budd-Chiari syndrome. Budd-Chiari syndrome is when you get a thrombosis in the hepatic vein, which then results in infarction of the liver. And the most common cause of Budd-Chiari syndrome is polycythemia vera. Another complication would be a flushed face. The blood can congest within blood vessels due to its hyperviscosity, resulting in a flushing of the face. And finally, these people get itching after bathing. And the reason they get itching is because along with the red blood cells, they also have extra mast cells. So one of the cells that's particularly increased in polycythemia vera are the mast cells. Now the mast cells are loaded with histamine, and it so happens that when patients bathe, that can actually set off the mast cells to release their histamine, which then results in itching after bathing. It's a very classic symptom, especially, again, on examinations. The treatment of polycythemia vera is to decrease the red blood cell mass, and we do that by phlebotomy. Without

treatment, death would usually occur within a one-year period. A second-line treatment is hydroxyurea. One of the important states that needs to be distinguished from polycythemia vera is reactive polycythemia. Basically, the idea here is that if we have an increased red blood cell mass, the increased red blood cell mass or count could be due to a reactive condition. For example, let's say that the patient has lung disease, and because they have lung disease, they have hypoxia. And so if they have hypoxia, the SAO₂ would be low, and as a result, erythropoietin production would be increased. And so if the patient has a reactive polycythemia due to lung disease, then we would expect that the SAO₂ would be low, and that the erythropoietin, which is driving that reactive polycythemia, would be increased. Another way that we could get a reactive polycythemia would be if we had a tumor that ectopically produced erythropoietin. And the tumor that does that is renal cell carcinoma. That's high yield. So a tumor, for example, renal cell carcinoma, could ectopically produce erythropoietin, and in that case, the erythropoietin would be high, which would be driving that high red blood cell count, but the SAO₂ would be normal because there is no hypoxemia in the patient. Of course, in polycythemia vera, there's no problem with oxygenation, and so the SAO₂ is normal. And the erythropoietin would actually be decreased due to a negative feedback. There are so many red blood cells that the erythropoietin levels are decreased. And so this is very important to understand how to distinguish polycythemia vera from reactive polycythemia. Another of the myelodysplastic syndromes is essential thrombocythemia. In this particular disease, there's a neoplastic proliferation, again, of mature myeloid cells. However, the key cells are platelets. Red blood cells would be up, and granulocytes would also be increased, but the key cell are the platelets, also associated with a JAK2 kinase mutation, very high yield to be familiar with. Here's what the blood smear would look like. In essential thrombocythemia, you've got a ton of platelets. Another circumstance that could result in a very similar pattern would be iron deficiency anemia. Iron deficiency anemia can also have a lot of platelets associated with it. And so when we have a lot of platelets, we want to think about the possibility of essential thrombocythemia, but in the back of our mind, we should also think about the possibility of iron deficiency anemia. The symptoms of essential thrombocythemia will either be increased risk of bleeding or thrombosis. Basically, either the platelets will not be functioning, giving you an increased risk of bleeding, or the platelets will be over-functioning, increasing the risk for thrombosis. Now, most of the myelodysplastic syndromes have the ability to eventually burn out, and they would result in marrow fibrosis. However, this disease rarely goes to marrow fibrosis. Most of the myeloproliferative disorders have the ability to develop into acute leukemia. This disease rarely develops into acute leukemia. What is particularly important is that there is no significant risk for hyperuricemia or gout. Now recall that in most of the myelodysplastic syndromes, there is an increased risk for hyperuricemia and gout because you're turning over a ton of cells. So

again, in Puvera, you're making a ton of red blood cells, those red blood cells will have their nucleus removed before they get put into the blood, And so that nucleus will have to be broken down, and if we're gonna break down nucleus, we're gonna go through the purine degradation pathway, and the end result of the purine degradation pathway would be uric acid, giving you hyperuricemia. However, in the case of essential thrombocythemia, remember that how do platelets develop? If this is the megakaryocyte, the way by which the platelets develop is that they are actually just pieces of megakaryocyte cytoplasm, and the platelet really doesn't have a nucleus. And so when these platelets are being overproduced, there is no excess turnover of nuclear material because it's just pieces of cytoplasm. And so there is no significant risk for hyperuricemia or gout. The fourth of the myelodysplastic syndromes is myelofibrosis. In this particular case, we get a neoplastic proliferation of mature myeloid cells, particularly the megakaryocytes, again, driven by a JAK2 kinase mutation. The megakaryocytes produce excess platelet-derived growth factor, and that platelet-derived growth factor then results in marrow fibrosis, which creates the classic myelofibrosis that we based the name of this disease on. Here's a picture highlighting the myelofibrosis. This is a spicula of bone, and we can clearly see that there's way too much fibrosis, pink collagen, deposited within this marrow space. All we should really expect to see in the marrow space is fat, and bone marrow elements. However, here we're seeing a ton of fibrosis, and so this is the classic finding in myelofibrosis on biopsy. Clinical features of myelofibrosis include splenomegaly. Now, the idea here is that once the bone marrow is fibrosed, hematopoiesis can no longer properly occur in the bone marrow, and so the stem cells basically remember or basically go back to the place where they used to make blood in the fetus, which was in the spleen or in the liver, resulting in extra hematopoiesis. Now at the same time, because there's massive extramedullary hematopoiesis, the patients develop something called a leukoerythroblastic smear. And the idea here is the following. When blood cells are made in the bone marrow, the bone marrow contains reticulin gates. What are the reticulin gates? They prohibit cells that are immature from actually exiting into the blood. So if these are the cells being developed over here, they won't exit into the blood until they're small enough to pass through these gates, which are basically made of a molecule called reticulin. In the spleen, when extramedullary hematopoiesis occurs, especially in massive amounts, We won't be able to prohibit immature cells from leaving the spleen, and so we get something called a leukoerythroblastic smear. Leuko, referring to the white cells, erythro, referring to the red cells, blastic, referring to an increased number of immature cells in the blood. In association with the leukoerythroblastic smear, we would also see nucleated red blood cells. The other finding that's important to note is that these patients will have an increased risk of infection, thrombosis, and bleeding. Now, why would they have an increased risk of infection, thrombosis, and bleeding? Well, basically, the bone marrow is no longer able to

properly produce your red blood cells, your white blood cells, and your platelets, and the vast majority of hematopoiesis has now shifted over to the spleen. However, the spleen is tiny compared to the entire bone marrow space, so the patients, although they are producing some extramedullary hematopoiesis, it's nowhere near what they actually need. And so there's an increased risk of infection, thrombosis, and bleeding. Here is a picture of a teardrop cell. And the reason why I included this in your text is to remind you that in myelofibrosis, there's also an increased risk of developing teardrop cells. The way I think about this is that although most of the hematopoiesis moves to the spleen, Some of the hematopoiesis does remain within the bone marrow, and you can imagine that if this is the bone marrow, and the bone marrow is totally fibrosed, and there's a little bit of hematopoiesis occurring, as the red blood cells are generated in this fibrosis, they'll be stretched as they try to leave to get out into the blood, and when they're stretched, they'll be squeezed into these teardrop shapes. And so this slide I included here to remind you of what a teardrop cell looks like and to also alert you to the fact that teardrop cells are commonly seen in myelofibrosis. And that concludes our discussion of myeloproliferative disorders.